California Grid Stress Monitoring Dashboard

Public EIA-930 grid operations analysis
across five California balancing authorities

Sileshi Hirpa

Data Science | Business Analytics | Python | Tableau

What problem does this project address?

California's electricity grid is managed by five balancing authorities that must match supply and demand in real time.

Analytical question:

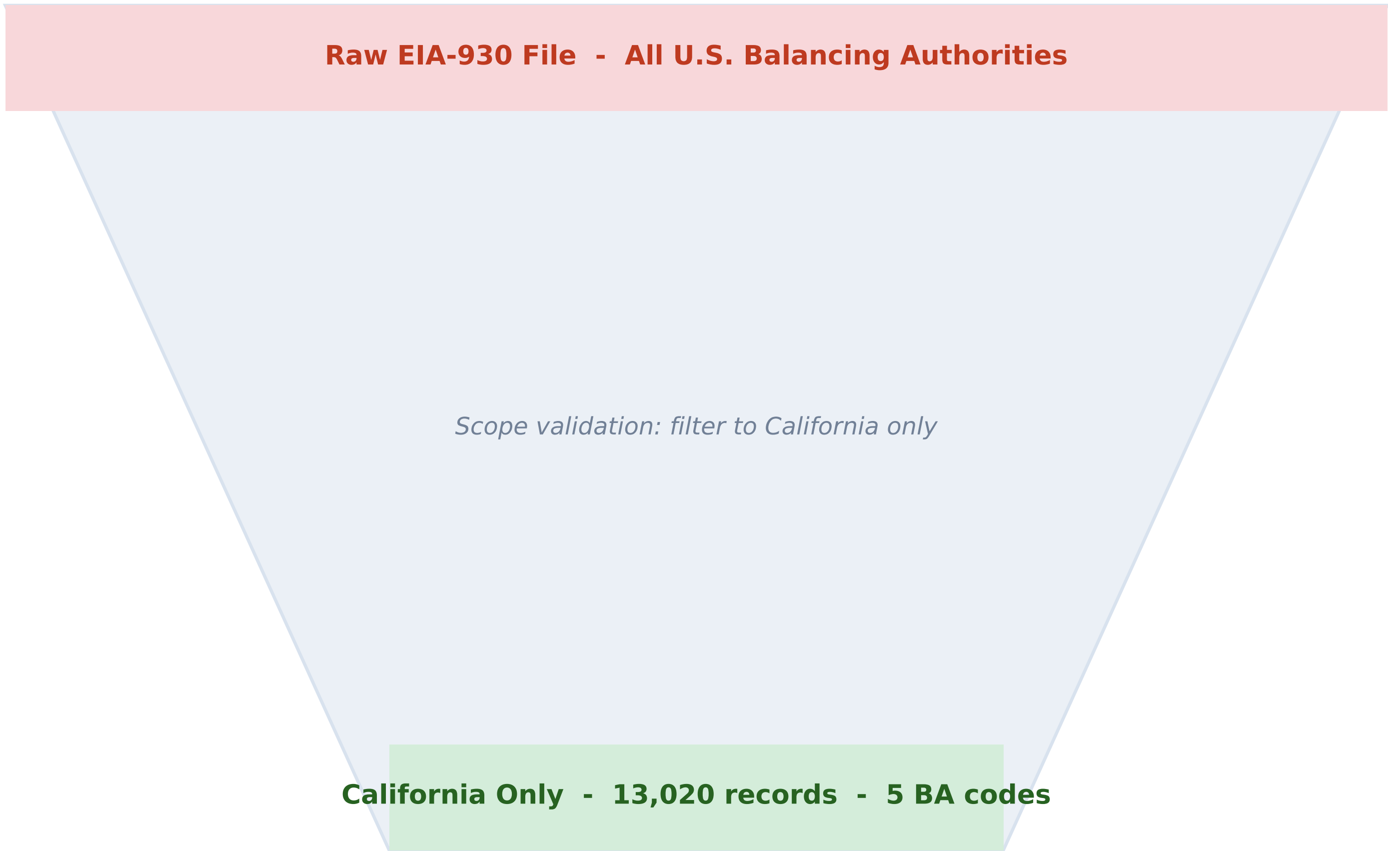
How can public hourly EIA-930 data be prepared, validated, and visualized so a reviewer can quickly identify periods that may deserve additional attention?

California Balancing Authorities in Scope

BANC	**CISO**	**IID**	**LDWP**	**TIDC**
Northern California	California ISO	Imperial Irrigation	Los Angeles DWP	Turlock Irrigation

> ***"A chart can run without error
> and still answer the wrong question."***

The raw EIA-930 file contains balancing authorities from across the United States.
An initial visualization ran without error but produced a misleading result
because non-California records were included. Scope was validated before analysis.



From raw CSV to published dashboard

PREPARE

- Load and inspect EIA-930 CSV
- Standardize column names
- Validate California scope
- Filter: 5 California BAs
- Convert UTC to Pacific Time

ENGINEER

- Compute forecast error
- Calculate generation gap
- Derive import pressure
- Build Grid Stress Index
- Classify review priority tiers

DELIVER

- Export 4 dashboard CSV files
- Build 3-tab Tableau workbook
- Publish to Tableau Public
- Document pipeline and limits
- Commit to GitHub

How the Grid Stress Index works

Stress Index = (Demand MW / Peak Demand MW) x 100

Normalized per authority against its own peak - enables fair comparison across scales.

Why relative normalization matters:

Both can reach Stress Index 90+ relative to their own peaks.

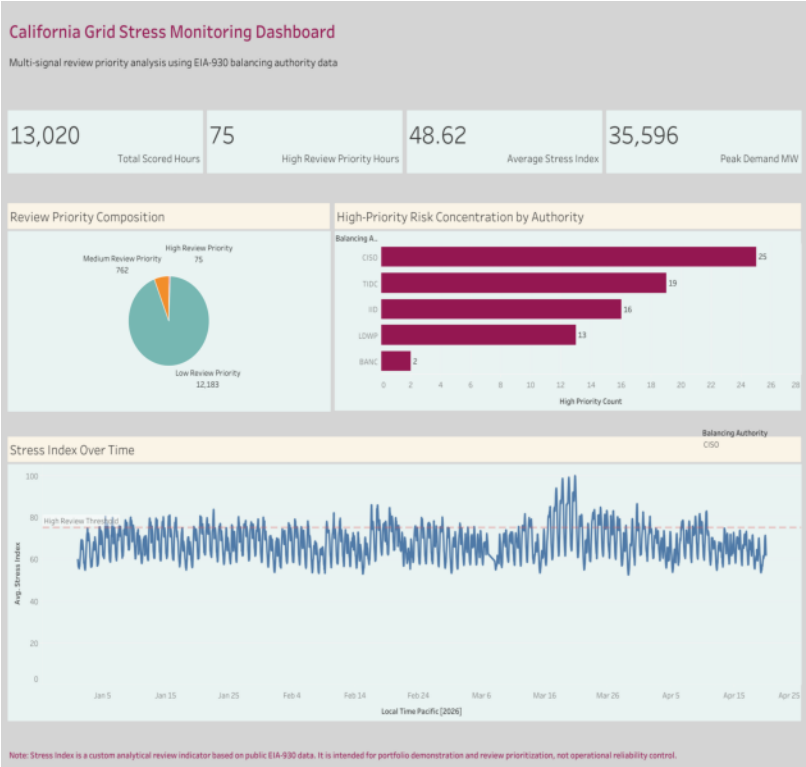
Review Priority Tiers

TIER	THRESHOLD	HOURS
High	Stress Index >= 90	75 hrs
Medium	75 <= Stress Index < 90	762 hrs
Low	Stress Index < 75	12,183 hrs

Note: peak denominator is based on the January-April 2026 dataset window, not a multi-year historical record.

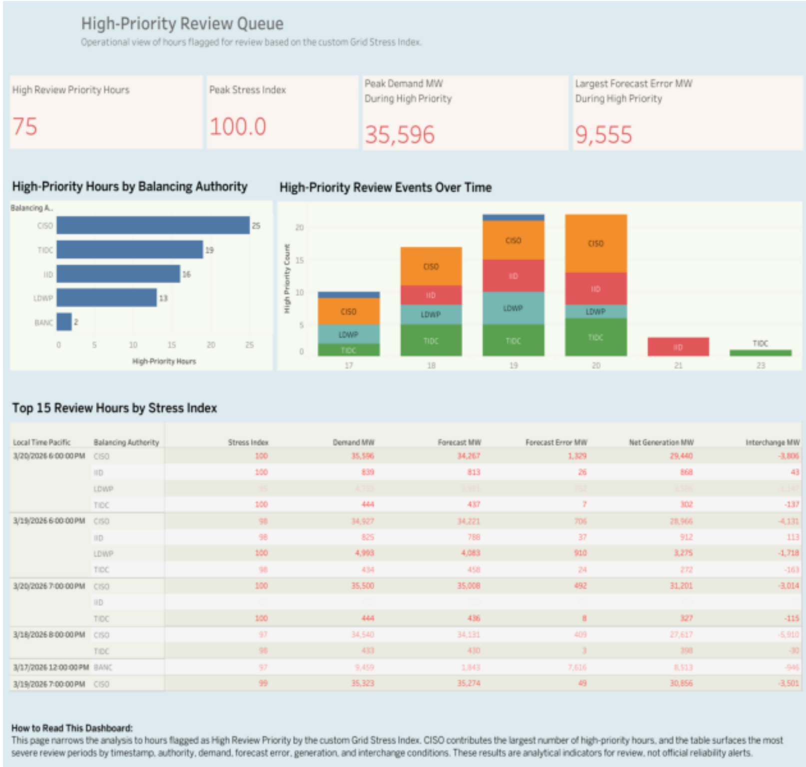
Published on Tableau Public

Executive Overview



KPI summary cards, priority tier composition, and Stress Index trend over the full Jan-Apr 2026 window. Designed for reviewers who need to orient quickly across all five balancing authorities.

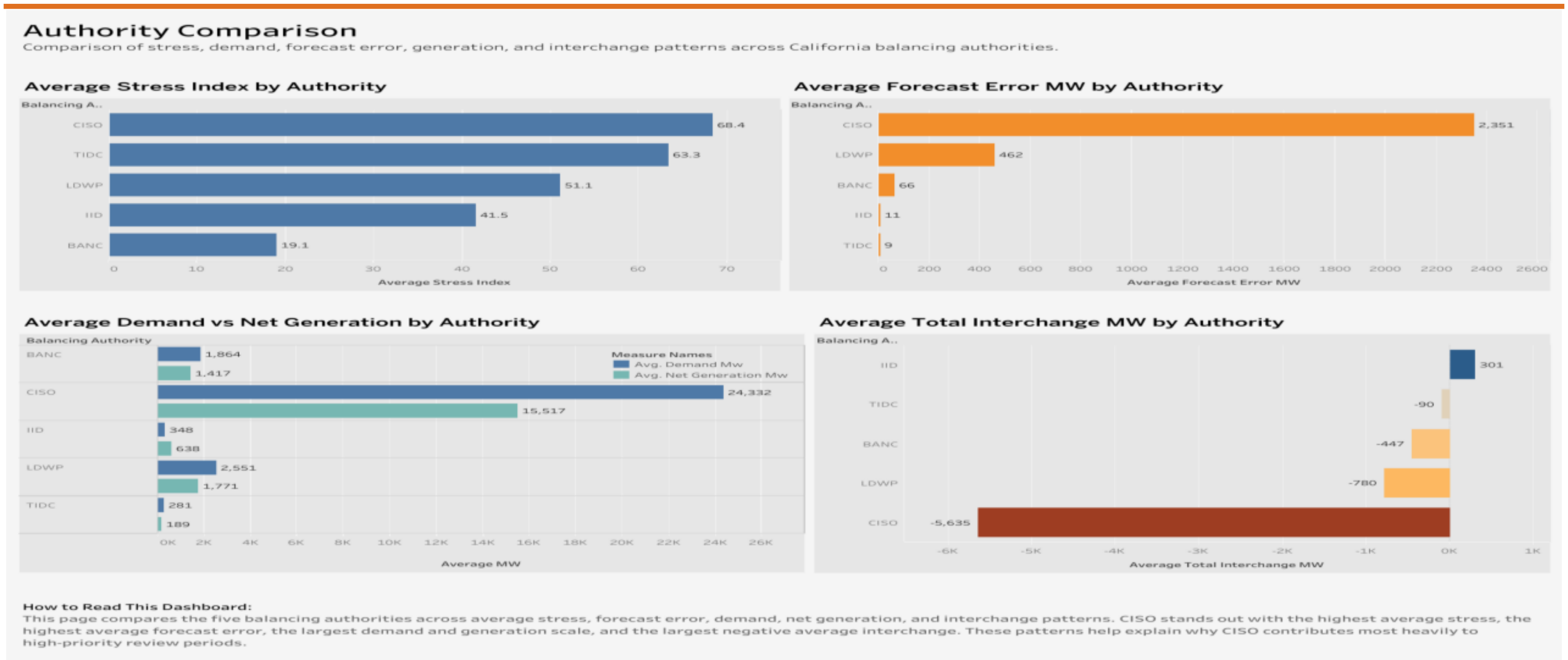
High-Priority Review Queue



The 75 High Review Priority hours ranked by Stress Index, with peak demand, forecast error, and authority code. Structured for record-by-record operational review and drill-down into flagged windows.

Authority Comparison and full-suite interpretation

Authority Comparison



Places BANC, CISO, IID, LDWP, and TIDC side by side across Stress Index, forecast error, demand vs. net generation, and total interchange.

Full Dashboard Suite - Big Picture

Together, the three tabs move a reviewer from grid-wide summary to specific high-priority hours to cross-authority pattern recognition.

One analytical question answered at three levels of resolution: overview, flagged events, and authority-by-authority comparison.

What the data showed

13,020 Total Scored Hours	**35,596 MW** Peak Demand (CISO)
75 High Review Priority Hours	**100.0** Peak Stress Index
48.62 Average Stress Index	**9,555 MW** Largest Forecast Error (High Priority)

What this project demonstrates

- Python - pandas, Plotly, matplotlib
- Feature engineering - custom metric design
- Time-series analysis across multiple authorities
- Dashboard data modeling - layered CSV exports
- Tableau Public - 3-tab interactive dashboard
- Scope validation and analytical communication
- GitHub documentation and reproducible pipelines
- Responsible interpretation of public data

Responsible Interpretation

The Grid Stress Index is a custom review indicator for this portfolio project.
It is not an official reliability metric or utility methodology.
All data is from public EIA-930 records published by the U.S. EIA.

Thoughts or questions? I would welcome feedback in the comments.

Full project and dashboard links in the first comment.

github.com/sileshith/california_grid_analysis